Section: ___________ Name:____________________________________

Read the following directions carefully. DO NOT turn to the next page until the exam has started.

Write your name and section number at the top right of this page:

Class	**Section Number**
Sahifa 8:50	001
Miranda 9:55	002
Sahifa 2:25	003
Sahifa 3:30	004
Miranda 8:50	005

As you complete the exam, write your initials at the top right of each other page.

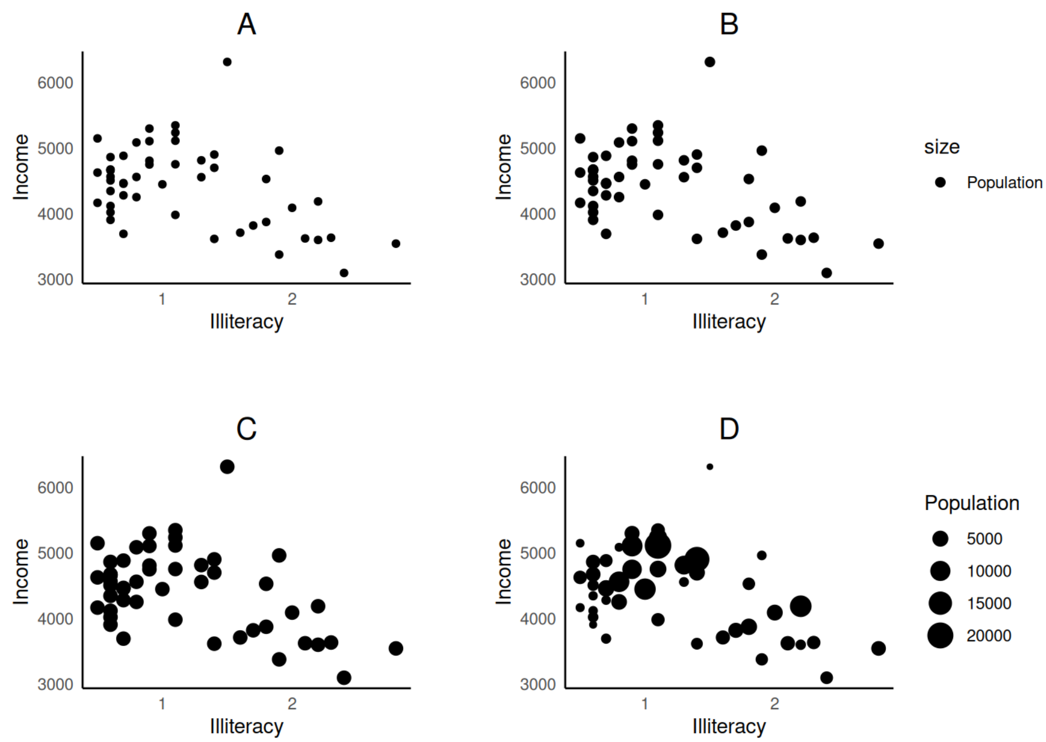
When the exam start time is called, you may turn the page and begin your exam.
If you need more room, there is a blank page at the end of the exam, or we can give you some scratch paper.

Some multiple choice questions are "Select ONE" while others are "Select ALL that apply". Pay attention to the question type and only mark one option if it says "Select ONE". Fill in the circles completely.

If you finish early, you can hand your exam to your instructor or TA and leave early.

Otherwise, stop writing and hand your exam to your instructor or TA when the exam stop time is called.

1. Below are four different `geom_point()` plots of illiteracy rate versus income for US states in the `states` dataframe. Some plots also show the states' population.



(a) Match the four plots (A, B, C, D) to the four different `geom_point()` calls below.

___A___ `geom_point(aes(x = Illiteracy, y = Income))`

___D___ `geom_point(aes(x = Illiteracy, y = Income, size = Population))`

___B___ `geom_point(aes(x = Illiteracy, y = Income, size = "Population"))`

___C___ `geom_point(aes(x = Illiteracy, y = Income), size = 3)`

(b) The following code returns an error when trying to run `geom_smooth()`. How should the error be fixed? **Select ONE**.

```
states %>%
  ggplot() +
  geom_point(aes(x = Income, y = Illiteracy)) +
  geom_smooth()
```

○ `geom_smooth()` should be added before `geom_point()`.
○ `x = Income` and `y = Illiteracy` should be moved outside of `aes()`.
● The `aes()` mapping should be moved inside of `ggplot()`.
○ None of the above; the code should not return any error.

2. A dataframe `lions` has 32 rows, each representing a unique lion.

   It has two numeric columns: `age`, each lion's age in decimal years, and `proportion.black`, the percentage of that lion's nose which is black. It has a logical column `adult`, which is `TRUE` if `age` is greater than 3 and `FALSE` otherwise.

   (a) Consider the following code which creates a new dataframe, `lions_summary`.

   ```
   lions_summary <- lions %>%
    group_by(adult) %>%
    summarize(averageAge = mean(age),
    averagePropBlack = mean(proportion.black))
   ```

   Which of the following statements are true about `lions_summary`? **Select ALL that apply**.

   - ● `lions_summary` has 2 rows.
   - ○ `lions_summary` has 2 columns.
   - ● `lions_summary` has a column called "`adult`".
   - ○ If there were any `NA` values in the `age` column of `lions`, all the values of `averageAge` in `lions_summary` will be `NA`.

   (b) Which lines of code can be used to keep only the rows of `lions` whose age is greater than 3? **Select ALL that apply**.

   - ● `lion %>% filter(adult)`
   - ○ `lion %>% filter(age)`
   - ● `lion %>% filter(age > 3)`
   - ○ `lion %>% select(adult)`
   - ○ `lion %>% select(age)`
   - ○ `lion %>% select(age > 3)`

   (c) The dataframe `lion_traits` has 25 rows, each representing a unique lion. It has three columns: `id`, a number that uniquely identifies each lion, `sex`, a categorical variable for whether the lion is male or female, and `length`, the length of the lion's body in meters.

   How many rows and columns will result from the following join operation? Briefly explain your answers. If the code returns an error, explain why.

   ```
   left_join(lions, lion_traits)
   ```

   The code will return an error. There are no columns in common between the two datasets, so there is nothing to join by.

3. The plot above shows the densities of three different distributions, named Solid, Dashed, and Dotted. One of these distributions is $N(0, 1)$, standard normal, another one is $t(5)$, the t distribution with 5 degrees of freedom, and the final one is $D(0, 1)$, some other distribution with mean 0 and standard deviation 1.

   (a) (3 points) Identify each distribution by the corresponding line type.

   - $N(0, 1)$: ● Solid ○ Dashed ○ Dotted
   - $t(5)$: ○ Solid ○ Dashed ● Dotted
   - $D(0, 1)$: ○ Solid ● Dashed ○ Dotted

   (b) Rank the three distributions using their line type names (Solid, Dashed, Dotted) in order from that with the smallest to the largest values of the 0.975 quantile.

   (Smallest) Dashed < Solid < Dotted (Largest)

   (c) A random sample of size $n = 400$ is drawn from distribution $D(0, 1)$ and we calculate the sample mean

   $$\bar{X} = \frac{1}{400} \sum_{i=1}^{400} X_i.$$

   What are the mean and standard deviation of the distribution of $\bar{X}$? **Select ONE.**

   ○ $\mu_{\bar{X}} = 0, \quad \sigma_{\bar{X}} = 1$
   ○ $\mu_{\bar{X}} = 0, \quad \sigma_{\bar{X}} = \frac{1}{400}$
   ○ $\mu_{\bar{X}} = 400, \quad \sigma_{\bar{X}} = 400$
   ● $\mu_{\bar{X}} = 0, \quad \sigma_{\bar{X}} = \frac{1}{\sqrt{400}}$

   (d) Which of the three distributions will have a shape closest to that of the distribution of $\bar{X}$? (Note that the scale will be different.) **Select ONE.**

   ● $N(0, 1)$ ○ $t(5)$ ○ $D(0, 1)$

4. A random viewer's ideal movie length, in minutes, is approximately $X \sim N(120, 15)$. Actual movie lengths are given by $Y \sim N(143, 19)$.

   (a) Which R code below calculates the probability that a random movie is shorter than the 40th percentile for ideal movie length? **Select ONE**.

   - ● `qnorm(0.4, 120, 15) %>% pnorm(143, 19)`
   - ○ `qnorm(0.4, 143, 19) %>% pnorm(120, 15)`
   - ○ `pnorm(0.4, 120, 15) %>% qnorm(143, 19)`
   - ○ `pnorm(0.4, 143, 19) %>% qnorm(120, 15)`

   (b) What movie length y corresponds to a z-score of 1? **Select ONE**.

   - ○ 19
   - ○ 124
   - ○ 143
   - ● 162

5. Define four different independent binomial random variables.

   $X_1 \sim \text{Binom}(5, 0.6)$, $X_2 \sim \text{Binom}(25, 0.4)$, $X_3 \sim \text{Binom}(100, 0.1)$, $X_4 \sim \text{Binom}(100, 0.4)$

   (a) Fill in the blanks below with "less than", "greater than", or "equal to".

   - The expectation of X_1 is ___greater than___ 2.5.
   - The smallest possible value of X_1 is ___equal to___ the smallest possible value of of X_4.
   - The expectation of X_3 is ___less than___ the expectation of X_4.
   - The standard deviation of X_3 is ___less than___ the standard deviation of X_4.

   (b) Which of the resulting combinations below is also a binomial random variable? **Select ONE**.

   - ○ $Y_1 = X_1 + X_2$
   - ○ $Y_2 = X_2 + X_3$
   - ● $Y_3 = X_2 + X_4$
   - ○ $Y_4 = X_3 + X_4$

   (c) Which of the four variables is *least* well approximated as $N\left(np, \sqrt{np(1-p)}\right)$? **Select ONE**.

   - ● X_1
   - ○ X_2
   - ○ X_3
   - ○ X_4

6. A regression line is fit on 50 (x, y) pairs. Summaries of the data and the regression line are given below.

	X	Y
Mean	26.8	80.1
Range	[10.5, 39.8]	[29.8, 123.4]

Regression line: $\hat{y}_i = 4.06 + 2.84x_i$

(a) A 95% prediction interval for y at $x = 50$ is reported as (129.15, 162.25). Why would this result be considered extrapolation? **Select ONE**.

- ● The value 50 is outside of the range of the original X values.
- ○ The lower bound of the interval is outside the range of the original Y values.
- ○ The interval fails to cover the mean of the original Y values.
- ○ None of the above; this is not considered extrapolation.

(b) A test for the linear regression slope against null hypothesis $H_0 : \beta_1 = 0$ results in a test statistic of $t_{obs} = 20.8$. What is the standard error of the estimated slope?

Using the formula for a slope test statistic, we have

$$\begin{aligned}\text{Test stat} &= \frac{\text{Estimated slope} - \text{Null}}{\text{Standard error}}\\ 20.8 &= \frac{2.84 - 0}{se}\\ se &= \frac{2.84}{20.8}\end{aligned}$$

(c) Based on the value of test statistic, which of the following conclusions is most reasonable? **Select ONE**.

- ○ X and Y have a negative relationship.
- ● X and Y have a positive relationship.
- ○ X and Y have no linear relationship relationship.
- ○ The value of X causes a significant change in the value of Y.

7. Data is collected on the duration each winter that a lake's surface is frozen over a period of 103 consecutive years. The correlation coefficient between the variables is $r = -0.4$.

   The year variable has a mean $\bar{x} = 1950$ and standard deviation $s_x = 30$. The freeze duration variable has $\bar{y} = 90$ and standard deviation $s_y = 17$. Consider fitting a simple linear regression model to this data.

   (a) Write a numerical expression sing only numbers and arithmetic symbols (like + or ×) for the predicted freeze duration in the year 1890. No need to simplify.

   $$\hat{\beta}_1 = -0.4\Big(\frac{17}{30}\Big), \qquad \hat{\beta}_0 = 90 - \Big(-0.4\Big(\frac{17}{30}\Big)\Big) \times 1950$$

   $$\text{Predicted 1890 duration} = \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1(1890)$$

   (b) A 90% confidence interval for the freeze duration of the true regression line at $x = 1890$ has the form:

   $$\hat{y}_1 \pm c \times SE_C$$

   A 90% prediction interval for the duration that the lake was frozen in the single year 1890 has the form

   $$\hat{y}_1 \pm c \times SE_P$$

   where c is a critical value from some distribution and SE_C, SE_P are some positive values. Which R code calculates the critical value c? **Select ONE**.

   ○ `qnorm(0.95)`
   ○ `qnorm(0.975)`
   ● `qt(0.95, df = 101)`
   ○ `qt(0.975, df = 101)`
   ○ `qt(0.95, df = 102)`
   ○ `qt(0.975, df = 102)`

   (c) Circle the true relationship between SE_C and SE_P. Briefly justify your choice.

   $SE_C < SE_P$ $\qquad$ $SE_C = SE_P$ $\qquad$ $SE_C > SE_P$

   A confidence interval for the regression line is narrower than the prediction interval at the same x. Since they have the same critical value, we must have $SE_C < SE_P$. The prediction standard error also takes into account the additional variability from a new data point.

8. From 1999 to 2008, the U.S. Mint released a huge quantity of special commemorative quarters (25 cent coins), each of which had a unique design on the back celebrating one of the fifty states. Let p be the true proportion of all the commemorative quarters that have a Wisconsin design on the back. Consider the following hypothesis test:

$$H_0 : p = 1/50 \text{ vs. } H_\alpha : p > 1/50$$

You randomly collect 200 of these commemorative quarters and find that 11 have Wisconsin designs on the back.

In conducting this test, you may assume that any relevant assumptions are met and do not have to mention nor explain them.

(a) What is the observed value of the test statistic for this test?

The test statistic is the count of "successes", which are Wisconsin quarters. The observed test statistic in this case is 11.

(b) Assuming the null hypothesis is true, what is the sampling distribution of the test statistic?

$$X \sim \underline{\text{Binom}} \left(\underline{200}, \underline{\tfrac{1}{50}} \right)$$

(c) Which line of R code correctly calculates the p-value for this test? **Select ONE**.

- ○ `pbinom(11, 200, 1/50)`
- ○ `pbinom(11-1, 200, 1/50)`
- ○ `1 - pbinom(11, 200, 1/50)`
- ● `1 - pbinom(11-1, 200, 1/50)`

(d) Assume the p-value is 0.003. **Select ALL** correct statements or justifiable conclusions

- ● The result is statistically significant at the $\alpha = 0.05$ level.
- ● There is strong evidence that the proportion of special commemorative state coins with Wisconsin on the back is greater than 1/50.
- ○ Based on the same data, a 95% confidence interval for the true proportion of Wisconsin quarters would contain the value 1/50.
- ○ The p-value from a two-sided test with the same data would also be equal to 0.003.

9. A 95% confidence interval for the difference in population proportions $p_1 - p_2$ is

$$-0.1 \ < \ p_1 - p_2 \ < \ 0.2.$$

(a) What is true about the p-value of a two-sided hypothesis test of $H_0 : p_1 = p_2$ versus $H_a : p_1 \neq p_2$? **Select ONE**.

○ The p-value is less than 0.05.
○ THe p-value is equal to 0.05.
● The p-value is greater than 0.05.

(b) The interval was created using the Agresti-Coffe adjustment. If X_1, X_2 and n_1, n_2 are the number of successes and the number of trials from each group, what is the A-C adjusted point estimate for $p_1 - p_2$? **Select ONE**.

○ $\dfrac{X_1}{n_1} - \dfrac{X_2}{n_2}$

● $\dfrac{X_1 + 1}{n_1 + 2} - \dfrac{X_2 + 1}{n_2 + 2}$

○ $\dfrac{X_1 + 2}{n_1 + 4} - \dfrac{X_2 + 2}{n_2 + 4}$

○ $\dfrac{X_1 + X_2 + 2}{n_1 + n_2 + 4}$

10. Identify each of the following scenarios as having independent or paired data.

(a) A teacher wants to compare the performance of 200 students on the practice SAT and actual SAT exam. Each student has their score recorded for both the practice and actual exam.

○ Independent ● Paired

(b) A teacher wants to compare the performance of students in her 9:00 AM class with students in her 1:00 PM class. She selects 100 students from each class and records the SAT score for each one.

● Independent ○ Paired

11. What lines of R code can be used to perform a paired T test on x and y? **Select ALL that apply**.

○ `t.test(x, y)` ● t.test(x, y, paired = T)
● t.test(x - y) ○ `t.test(x - y, paired = T)`

12. In a large bookstore, you are trying to determine if the average number of pages of a book in the sci-fi section is *different* from the average number of pages of a book in the mystery section.

- Let μ_S be the average page length of all books in the sci-fi section.
- Let μ_M be the average page length of all books in the mystery section.

(a) State the appropriate null and alternative hypotheses for this test.

We want to know if the average is different, which implies a two-sided test for a difference in means.

$$H_0 : \mu_S - \mu_M = 0 \quad \text{versus} \quad H_A : \mu_S - \mu_M \neq 0$$

(b) You take a random sample of 30 sci-fi books and find the average page length to be 236.4 with standard deviation 23.8. You take a random sample of 40 mystery books and find the average page length to be 231.3 with standard deviation 52.6.

Write a numerical expression that calculates the test statistic for your hypotheses. Do not simplify, evaluate, or round.

We are using the Welch T test statistic for a difference in means.

$$t_{obs} = \frac{\bar{x}_S - \bar{x}_M - 0}{\sqrt{\frac{s_S^2}{n_S} + \frac{s_M^2}{n_M}}} = \frac{236.4 - 231.3 - 0}{\sqrt{\frac{23.8^2}{30} + \frac{52.6^2}{40}}}$$

(c) Consider the following output of `t.test()` on this data. **Select ALL** true statements or justifiable conclusions.

```
        Welch Two Sample t-test

data:  scifi and mystery
t = 0.54033, df = 57.381, p-value = 0.5911
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -13.72092  23.86401
sample estimates:
mean of x mean of y
 236.3856  231.3140
```

- ○ There is strong evidence that the population mean length of sci fi books is larger than that of mystery books.
- ● The sample mean length of sci fi books is greater than the sample mean of mystery books.
- ○ The two-sided hypothesis test is significant at the $\alpha = 0.05$ level.
- ● There is a lack of evidence that the population mean lengths of either book genre differ from each other.